ZIRUO LI



Agricultural Automation · Computer Vision · Intelligent Robotics

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EDUCATION

China Agricultural University | M.Eng. in Agricultural Electrification and Automation · GPA 3.64 — **Sep 2023 - Jun 2026**

East China Jiaotong University | B.Eng. in Railway Traffic Signal and Control · GPA 3.61 · Top 10% — **Sep 2018 - Jul 2022**

RESEARCH EXPERIENCE

Shanghai Jiao Tong University | Institute of Mechatronics & Logistics Equipment · — **Jun 2026 - Present**

- Autonomous sweet pepper harvesting with embodied intelligence: integrated visual perception and robot control for target detection, localization, and autonomous manipulation under severe occlusion in greenhouse environments.
- VLA-based decision making for harvesting robots: introduced a vision-language-action model to combine visual observations and language instructions for task understanding and action planning.

China Agricultural University | Key Laboratory of Smart Agriculture System Integration Research, Ministry of Education · Advisor: Prof. Gang Liu — **Sep 2023 - Jun 2026**

- Developed a dual-stream ResNet50 model that fused top-view and side-view RGB features for non-contact beef cattle body-weight estimation.
- Built a keypoint-based pipeline for automatic cattle body measurement, including body height and body length extraction from images.
- Proposed a lightweight YOLOv5 model for individual recognition in group-housed beef cattle, achieving 93.2% precision while reducing parameters by 52.1% for edge deployment.

East China Jiaotong University | Signal Innovation Laboratory · Advisor: Assoc. Prof. Xiaoming Wang — **Nov 2018 - Mar 2021**

- Designed an RFID-based handheld railway inspection platform supporting data acquisition, precise positioning, anomaly alerts, and task feedback.
- Developed a microcontroller-based track inspection system using ultrasonic and temperature sensors, Wi-Fi transmission, real-time PC monitoring, and gauge anomaly alarms.

INDUSTRY EXPERIENCE

China Railway Electrification Engineering Group | Signaling Engineer · Signaling and Communications Design Institute — **Aug 2022 - Jun 2023**

- Supported railway signaling and communications system design, technical research, equipment selection, installation, commissioning, and on-site implementation.
- Participated in integrated testing for Xi'an Metro Line 16, including overspeed stopping, SKPS-triggered stopping, train-door/platform-screen-door coordination, ATC functions, ATO travel time, and joint power-on self-tests.

SELECTED FUNDED AND INDUSTRY PROJECTS

- **Spatiotemporal stability mechanisms of dairy-cow gait and early lameness recognition** - Technical Member; National Key R&D Program of China (14th Five-Year Plan); Jun 2023 - Dec 2024
- **Automated monitoring technologies and equipment for livestock physiological traits** - Technical Member; National Key R&D Program of China (14th Five-Year Plan); May 2024 - Mar 2026
- **Key technologies for intelligent sweet pepper harvesting robots in protected cultivation** - Technical Member; National Key R&D Program of China; May 2026 - Present
- **Computer-vision-based beef cattle weight estimation** - Technical Lead; JD.com; Nov 2023 - Nov 2024
- **Non-contact beef cattle body measurement system** - Project Lead; Zhengzhou Qifeite Technology Co., Ltd.; Nov 2024 - Jul 2025

PUBLICATIONS

Five first-author papers: two SCI-indexed and three EI-indexed. Asterisks denote corresponding authors.

1. **Ziruo Li**, Yadan Zhang, Xi Kang, Tianci Mao, Yanbin Li, and Gang Liu*. Individual Recognition of a Group Beef Cattle Based on Improved YOLO v5. Agriculture, 2025, 15(13): 1391. https://doi.org/10.3390/agriculture15131391. **[Published · SCI]**
2. **Ziruo Li**, Yadan Zhang, Yao Chong, Ying Han, Zenglong Song, Xueting Zeng, Xiaocong Li, and Gang Liu*. Beef Cattle Body Weight Estimation Based on Dual-View RGB Images. Animals, 2026, 16(16): 2532. https://doi.org/10.3390/ani16162532. **[Published · SCI]**
3. **Ziruo Li**, Xiaocong Li, Yadan Zhang, Xueting Zeng, Wenlong Yin, Chong Yao, and Gang Liu*. Automatic Measurement of Beef Cattle Body Dimensions Based on Improved YOLO11n-Pose and Effective Pose-Frame Screening. Transactions of the Chinese Society for Agricultural Machinery. Online first. https://link.cnki.net/urlid/11.1964.S.20260731.1003.002. **[Published · EI]**

4. **Ziruo Li**, Zenglong Song, Wenlong Yin, and Gang Liu*. Automatic Measurement of Beef Cattle Body Dimensions Based on Improved YOLO11n-Pose. Transactions of the Chinese Society for Agricultural Machinery. **[Accepted · EI]**
5. **Ziruo Li**, Xueting Zeng, Yadan Zhang, Wenlong Yin, Xingdan Wang, and Gang Liu*. Beef Cattle Body Weight Estimation under Natural Walking Conditions Based on a Cross-Attention Mechanism. Transactions of the Chinese Society for Agricultural Machinery. **[Accepted · EI]**
6. Mengyuan Chu, Yongsheng Si, **Ziruo Li**, Qian Li, and Gang Liu*. Multi-feature image layers fusion for accurate detection of dairy cow mastitis using deep learning. Computers and Electronics in Agriculture, 2025, 239: 110937. https://doi.org/10.1016/j.compag.2025.110937. **[Published · SCI]**
7. Yadan Zhang, Xueting Zeng, **Ziruo Li**, Yize Liu, Xiaocong Li, Ying Han, Gang Liu, and Jun Wang. Research on an automated detection for sheep estrus based on multimodal fusion of thermal images and sensor data. Computers and Electronics in Agriculture, 2026, 249: 111848. https://doi.org/10.1016/j.compag.2026.111848. **[Published · SCI]**
8. Ying Han, Yongsheng Si, Zhijiang He, Qian Li, **Ziruo Li**, Man Zhang, and Gang Liu*. Dynamic weighing system for dairy cows based on arrayed weighing platforms. Computers and Electronics in Agriculture, 2025, 230: 109943. https://doi.org/10.1016/j.compag.2025.109943. **[Published · SCI]**
9. Chong Yao, Qian Li, **Ziruo Li**, Tianci Mao, Xue Tian, Ping Song, Miao Zhang*, and Gang Liu*. Optimizing Two-Stream Asymmetric Feature Extraction Architecture with Dynamic Re-parameterization for Enhancing Beef Cattle Weight Estimation. Computers and Electronics in Agriculture. **[Published · SCI]**
10. Ping Song, Chong Yao, **Ziruo Li**, Tianci Mao, Ying Yang*, and Gang Liu*. Body Mass Estimation of Group-Housed Beef Cattle Based on Attention Mechanisms and Cross-Modal Hierarchical Feature Fusion. Transactions of the Chinese Society of Agricultural Engineering, 2025, 41(10): 221-231. https://doi.org/10.11975/j.issn.1002-6819.202412117. **[Published · EI]**

RESEARCH SOFTWARE AND INTELLECTUAL PROPERTY

- Image-Processing-Based Beef Cattle Weight Estimation System
- Deep-Learning-Based Beef Cattle Instance Segmentation System
- Intelligent Body Measurement Regression Analysis System
- Intelligent Cattle Image Processing System
- Smart Farm Visualization System Based on Spring Boot and Vue
- Upper-Computer Control Software for a Segmented Weighing System
- Cattle Keypoint Detection System
- Automated Infrared Eye-Temperature Detection System for Sheep

HONORS AND AWARDS

- Outstanding Graduate of Beijing Municipality
- Outstanding Graduate, China Agricultural University
- BYD Scholarship, 2024-2025 (13 recipients college-wide)
- First-Class Academic Scholarship, China Agricultural University, 2024-2025
- Second-Class Academic Scholarship, China Agricultural University, 2023-2024
- Second Prize, A5+N Artificial Intelligence and Remote Sensing Competition, 2025
- Outstanding Communist Youth League Member, China Agricultural University, 2024-2025
- Third Prize (Provincial), May Day Cup Mathematical Modeling Competition, 2024
- Outstanding Graduation Project, East China Jiaotong University, 2022
- Outstanding Student Leader, East China Jiaotong University, 2022
- Outstanding Youth League Branch, East China Jiaotong University, 2022
- Second Prize, Transportation Intelligent Networking Innovation Design Competition, 2019
- First Prize, 5th Green Building Intelligence Competition, 2019

TECHNICAL SKILLS

- Programming: Python and C++.
- Research methods: computer vision, deep learning, multimodal feature fusion, keypoint detection, pose estimation, instance segmentation, robotic perception, and edge deployment.
- Systems and applications: intelligent livestock phenotyping, agricultural robotics, VLA-based decision making, sensor integration, and real-time monitoring.

ENGLISH PROFICIENCY

- IELTS overall score: 6.5 · Listening 7.0 · Reading 6.5 · Writing 6.0 · Speaking 6.0

PROFESSIONAL PROFILE

- Strong foundation in mathematics and physics; experienced in reading and writing technical literature in English.
- Independent and analytical problem solver with a responsible work style, steady execution, and effective teamwork and organization skills.